Physics 3200Y: Lab 4

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Abstract

1 Introduction

This lab seeks to develop and test theoretical models for the field between two current carrying loops of wire. The developed models predict a high uniformity of the magnetic field between the coils in both magnitude and direction. The theoretical uniformity of the field will be investigated through the construction of the structure for which models were derived and the use of a magnetic field probe. Additionally the more mathematically challenging off-axis field of the coil structure will be measured and investigated.

2 Theory

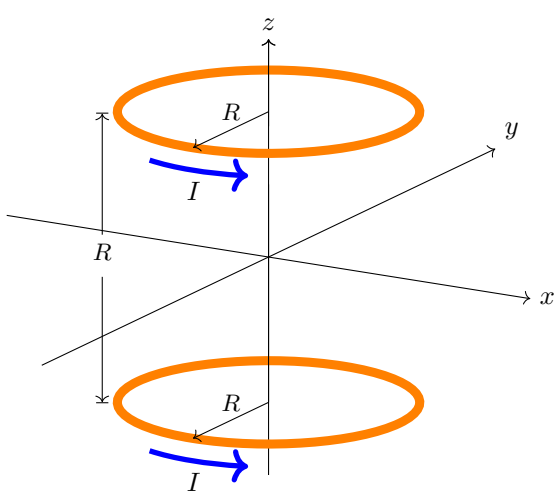

Figure 1: Helmholtz coil structure. Further reference to direction will be in terms of the $\parallel$ direction, the xy plane in the figure, and the $\perp$ direction, $\widehat{\mathbf{z}}$ in the figure.

The construction under study, known as Helmholtz coils is illustrated in fig. 3. We consider two coils, each of radius R, separated by a distance R. In the figure, and for the following derivation, the coils are placed at $\pm R/2$ respectively with the origin between them as indicated in the figure. We begin by deriving the field due to a single current carrying loop and then utilize linear superposition to determine the total magnetic field due to a true Helmholtz coil system. The magnetic field for a wire loop represented by a (directed) contour $\mathcal{C}$ is given by the Biot-Savart law,

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \int_{\mathcal{C}} \frac{d\boldsymbol{\ell}' \times \boldsymbol{\mathscr{r}}'}{\mathscr{r}^3} \tag{1}$$

where $\boldsymbol{\mathscr{r}} = \mathbf{r} - \mathbf{r}'$ and $\mathbf{r}$ and $\mathbf{r}'$ represent the test-point and an integration variable respectively. To exploit the symmetries present in the problem we consider only the field on the axis of symmetry, hereon referred to as the on-axis field. This means that our position vector $\mathbf{r} = (0, 0, z)$ in cylindrical coordinates. Since we are considering only the field due to a single coil we take $\mathbf{r}' = R\hat{\mathbf{s}}$ which sweeps out the entire coil. This gives $\boldsymbol{\mathscr{r}} = z\widehat{\mathbf{z}} - R\hat{\mathbf{s}}$. Finally if we direct the contour such that the area vector is in $\widehat{\mathbf{z}}$, $d\ell' = Rd\phi'\hat{\boldsymbol{\phi}}$. This means that eq. (1) becomes

$$\begin{aligned}\mathbf{B}(\mathbf{r}) &= \frac{\mu_0 I}{4\pi} \int_0^{2\pi} \frac{Rd\phi' z\hat{\mathbf{s}} + R^2 d\phi'\widehat{\mathbf{z}}}{(z^2+R^2)^{3/2}} \\ &= \frac{\mu_0 I}{2} \frac{R' z\hat{\mathbf{s}} + R^2\widehat{\mathbf{z}}}{(z^2+R^2)^{3/2}}\end{aligned}$$

and since the magnetic field obeys linear superposition, N such coils very close together create a field given by

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 N I}{2} \frac{R' z\hat{\mathbf{s}} + R^2\widehat{\mathbf{z}}}{(z^2+R^2)^{3/2}}. \tag{2}$$

Now we can construct the field due to two such coils at $\pm R/2$ easily again by linear superposition,

$$\begin{aligned}\mathbf{B}(\mathbf{r}) = \frac{\mu_0 N I R^2}{2} \Bigg[&\frac{1}{((z+R/2)^2+R^2)^{3/2}} \\ &+ \frac{1}{((z-R/2)^2+R^2)^{3/2}} \Bigg] \widehat{\mathbf{z}}\end{aligned} \tag{3}$$

where we've canceled out the $\hat{\mathbf{s}}$ components of the field by symmetry. We can plot this expression (in natural units), and we see that indeed the field inside and near the center is highly uniform. This is the desired effect of the Helmholtz configuration.

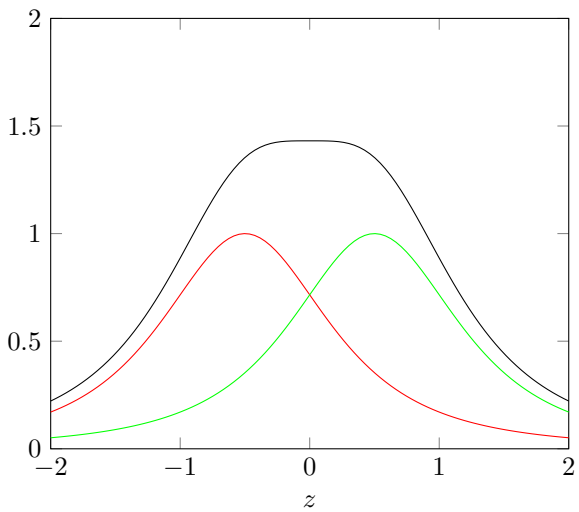

Figure 2: Helmholtz coil relative magnetic field strength. The coloured curves are fields due to the individual coils, placed at $\pm 1/2$, and the black curve is their sum.

3 Methods

3.1 Coils

Two 200 turn, 10 cm radius, Pasco Scientific Helmholtz coils were used [1]. These were mounted on a ruled base, also manufactured by Pasco Scientific. A DC current of approximately (not exceeding) 1.5 A was passed through the coils in series with the coils connected such that the field produced by each was in the same direction.

3.2 Magnetic Probe

The magnetic field produced by the coils was measured using a Pasco Scientific 2-axis field sensor [2]. The probe was positioned to measure the fields both perpendicular and parallel to the coil plane to verify the uniformity and self-canceling behaviour in these directions. Data was collected on the field in several configurations in order to verify the validity of assumptions made in theoretical derivations. Further discussion is available in the results section.

3.3 Linear Motion Sensor

In order to measure the field as a function of position in the coil a Pasco Scientific rotary motion sensor [3] was used, with the magnetic field probe mounted to a rigid support whose extension into the coils could be tracked by the motion sensor. Both the motion sensor and magnetic field probe were connected, via Pasco Scientific USB interfaces, to a desktop computer running the Pasco Capstone software for data collection.

3.4 Alignment

A challenge in configuring the experiment was the need to measure as closely as possible to the axis of symmetry of the coils, as this is the region in which the theory derived previously applies. To align the probe vertically two rulers were placed perpendicularly and the axial field indicator dot on the probe was aligned with the appropriate position on the rulers. To align horizontally a similar process was used except with the perpendicular field indicator dot. The motion sensor was zeroed at the midpoint between the coils.

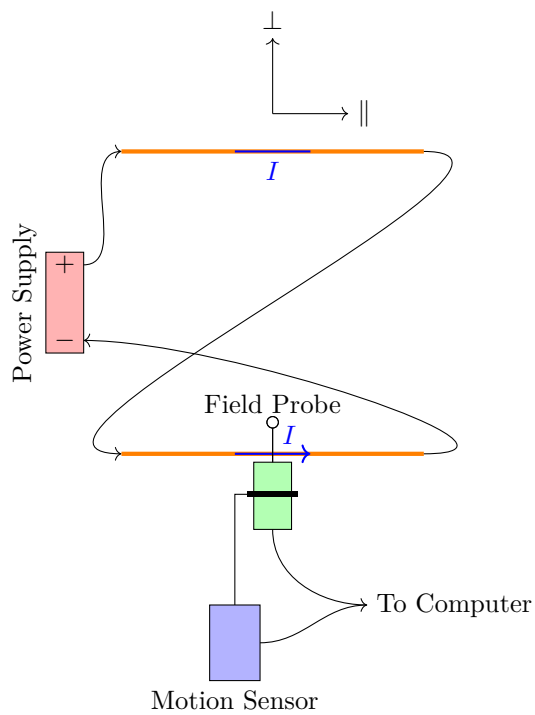

Figure 3: A cartoon of the experimental setup.

4 Results

4.1 On-Axis

The on-axis field was collected in several configurations. These are: the normal Helmholtz configuration, one coil on and the other off for each coil, both coils at a separation of $2R$, and both coils at a separation of $R/2$. For each run the current passing through the coils was held at a value not exceeding 1.5 A, though due to coil heating a drop to as low as 1.47 A was observed.

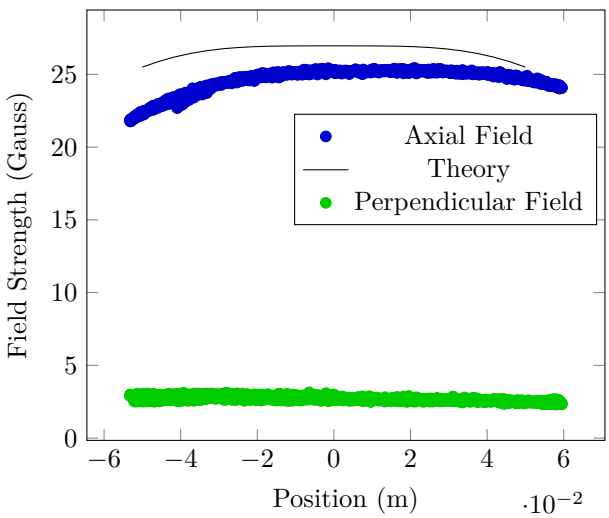

Figure 4: Perpendicular and axial fields in a standard Helmholtz configuration.

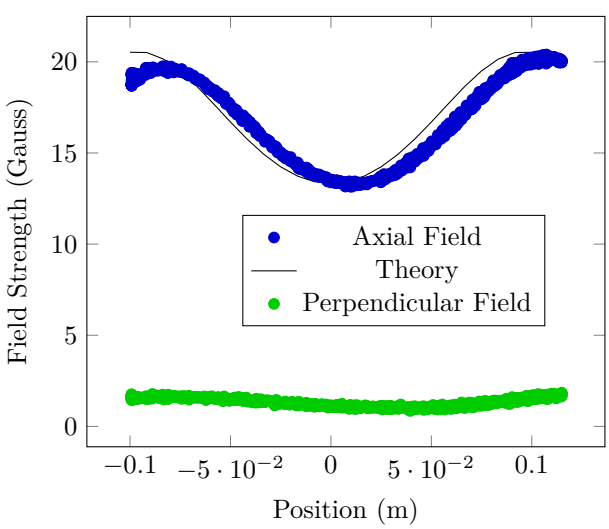

Figure 5: Perpendicular and axial fields in a $2R$ Helmholtz configuration.

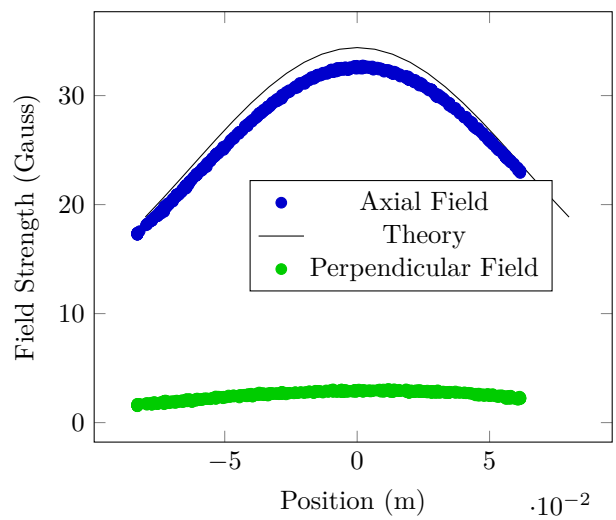

Figure 6: Perpendicular and axial fields in an $R/2$ Helmholtz configuration.

4.1.1 Superposition

Figure out a good way to

4.2 Off-Axis

Off axis field measurements were conducted at increasing (towards coil center) distance from the edge of the coils.

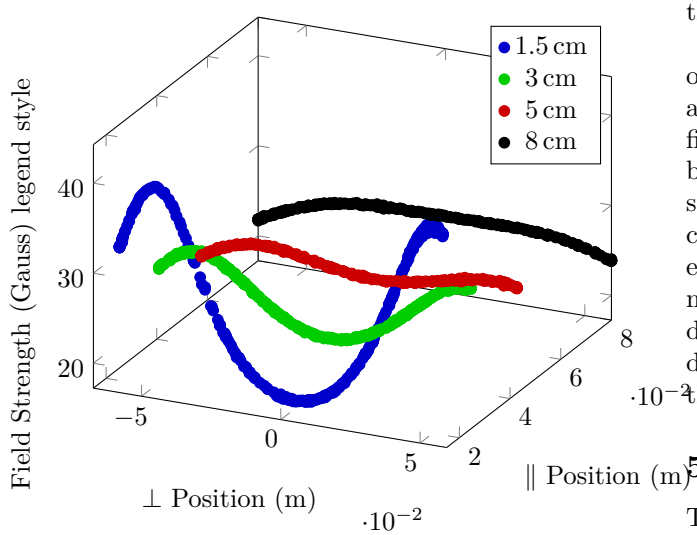

Figure 7: Axial field at varying distance from coil edge.

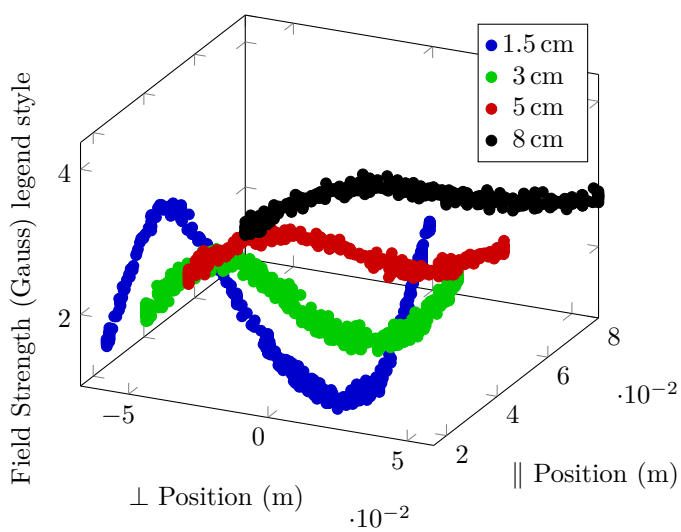

Figure 8: Perpendicular field at varying distance from coil edge.

5 Discussion

5.1 On-Axis

As can be seen in fig. 4, the developed theoretical model agrees somewhat with the collected data. The model estimates a higher overall axial field in the uniform and non-uniform regions. Additionally there is a noticeable horizontal shift between the model and data. This is likely due to inaccurate zeroing of the linear motion sensor when collecting data. Shifting the data by approximately 2 cm to the left provides much better alignment with the theoretical model. The collected perpendicular data agrees well with the assumption that the perpendicular field is very small, approximately an order of magnitude smaller than the axial field.

In fig. 5 and fig. 6 we see the expected nonuniformity of an improperly constructed Helmholtz coil system. We also see more variation in the perpendicular field. These figures both show the real accuracy of the model much better in the horizontal alignment. Interestingly they also show a much more similar magnetic field magnitude when compared to fig. 4. This points either to there being some effect which causes a field strength increase to match the model when in a non Helmholtz configuration or that the data in fig. 4 is inaccurate. It is more likely that the data is inaccurate, perhaps due to a lower current (the theoretical curves assume exactly 1.5 A).

5.2 Off-Axis

The off-axis measurements do not have a theoretical result to compare against, though fig. 8 does show the intuitively expected increase in perpendicular field strength due to closer proximity to one side of the coils. fig. 7 also shows a similar effect, again due to the proximity to one side of the coils removing the "balancing" of the opposite side of the coils. The effect becomes less present closer to the center of the coils as expected, quickly converging to a highly uniform field.

References

[1] Helmholtz coils • EM-6722 — pasco.com. `https://www.pasco.com/products/lab-apparatus/electricity-and-magnetism/coils-and-cores/helmholtz-coils`. [Accessed 30-03-2026].

[2] PASPORT 2-Axis Magnetic Field Sensor • PS-2162 — pasco.com. `https://www.pasco.com/products/sensors/pasport/pasport-2-axis-magnetic-field-sensor`. [Accessed 30-03-2026].

[3] PASPORT Rotary Motion Sensor • PS-2120A — pasco.com. `https://www.pasco.com/products/sensors/pasport/pasport-rotary-motion-sensor`. [Accessed 30-03-2026].